

Custom Sports Analytics Platform

SYSTEM ARCHITECTURE & DEPLOYMENT BLUEPRINT

This document summarizes our technical consultations regarding building a proprietary, AI-powered sports analytics system. The system specializes in real-time computer vision tasks, specifically **ball tracking** and **player posture/pose estimation**, capturing live video on-court and streaming it for backend processing.

1. Court-Side Connectivity Requirements (The 50m Bridge)

A standard camera setup cannot natively transfer high-fidelity video over 50 meters to a control room due to rapid transmission signal degradation. Two physical routing paths exist:

- **Option A (Network Approach - Recommended):** An IP or NDI-enabled camera connected via standard **Cat6 Ethernet cable**. This setup requires a **PoE (Power over Ethernet) Switch** in the control room, providing power and digital signal down a single line without needing court-side electrical outlets.
- **Option B (Broadcast Approach):** A camera outputting **SDI** over professional coaxial cabling, terminating at an **SDI Capture Card / USB Dongle** plugged directly into the control room PC.

2. Display & Modem Requirements

- **Screens:** No monitor or screen is required at court-level. However, the Control Room PC requires a high-resolution display monitor so local operators can manage settings, configurations, and baseline software workflows.
- **Modem:** A modem and hardwired internet connection are mandatory if video processing happens off-site (Cloud or Private Office). Wireless communication (Wi-Fi) must be strictly avoided to eliminate packet jitter and signal interruptions.

3. Choosing the Streaming Protocol for Live AI

Computer Vision pipelines require a flawless, high-fidelity frame progression. Dropped frames disrupt positional object vectors, while blur or pixelation breaks human skeletal structural mappings.

Recommended Protocol: SRT (Secure Reliable Transport)

SRT uses intelligent packet recovery (ARQ) over UDP. If data packets are dropped over the modem connection, SRT selectively retransmits only the missing data seamlessly. This prevents structural freezing while retaining exact image resolution and supporting efficient compression profiles (H.265/AV1).

Why alternatives fail: **RTMP** relies on strict TCP, meaning a network anomaly pauses the stream globally, forcing chronological data jumps. **WebRTC** sacrifices visual acuity (introducing blur/pixelation) to prioritize human latency speeds, destroying mathematical edge textures required by AI.

4. Critical Local Encoder Configuration

The control room PC must ingest the camera feed and compress it precisely to suit the computer vision backend requirements:

- **Framerate:** Minimum 60 FPS. High velocity sports components (balls/joints) create visual ghosting and trail artifacts at 30 FPS.
- **Bitrate Control:** Constant Bitrate (CBR) only, keeping image metrics rigid during fast motion sequences.
- **Keyframe Interval (GOP):** Set strictly to 1 Second. This prevents inter-frame "smearing" across predictive P-frames, feeding complete architectural snapshots to the model regularly.
- **B-Frames:** Set to 0 to prevent downstream computational delays and sequence manipulation.

5. Deployment Architecture Comparison

To implement your custom tracking model, three physical architectures are possible. Below is a detailed cost and operational breakdown:

Criteria	Pure Edge Node	Pure Cloud Ingest	Hybrid Office Hub (Private Cloud)
On-Court Cost	High (\$2,200+ for local AI GPU Rig)	Low (~\$700 basic encoder PC)	Low (~\$700 basic encoder PC)
Processing Cost	\$0 (Free, local processing)	High (Ongoing GPU hourly cloud rental)	\$0 (Free, centralized office hardware)
Internet Dependency	Low (Only sends analytics data dashboards)	Critical (Must sustain massive real-time streams)	Critical (Dual venue and office lines must stay live)
Security & Safety	Risky (Expensive hardware on-site)	Secure (Hosted in cloud data center)	Secure (Safely stored in your locked office)
Code Maintenance	Difficult (Requires onsite/remote updates)	Easy (Centralized code deployments)	Easy (Hardware sits right next to development team)

6. Deep Dive: The Hybrid Centralized Office Architecture

By positioning your primary hardware infrastructure inside your local corporate office, you create a private processing hub. The court PC acts strictly as a lightweight streaming node sending data down to your office infrastructure.

[COURT VENUE] [YOUR LOCAL OFFICE] Court Camera (Raw Feed) Office Router (Static IP / Port Forward) | | ▼ (50-meter Cat6 / SDI) ▼ Control Room PC (SRT Encoder) —[Public Internet]—> Office AI Workstation (NVIDIA GPU)

Infrastructure Blueprint for Office Implementation:

- **Static IP & Network Routing:** Your office internet provider must assign a permanent Static IP. Your network admin will configure local Port Forwarding (UDP port) or establish a network-level VPN (such as WireGuard) to route inbound venue video securely to the AI machine.
- **Aggregated Download Bandwidth:** Your office line must maintain substantial download headroom. While one court requires 12–15 Mbps, a cluster of 5 concurrent streaming courts requires a dedicated download buffer of at least 60–75 Mbps.
- **Hardware Pipeline Pipeline (FFmpeg + CUDA):** To prevent CPU bottlenecks, the office AI machine must utilize a media proxy (e.g., SRS or Nimble Streamer) coupled with specialized pipelines (GStreamer/FFmpeg with **NVDEC CUDA** integration). This translates the inbound SRT network stream directly into raw frames inside GPU VRAM, allowing PyTorch/YOLO/MediaPipe models to execute matrix evaluations without latency-inducing memory transfers.